06-10 第二篇

The Engineering Behind Bridges

Bridges connect places that nature has separated—rivers, valleys, and bays. They have been essential to human trade, travel, and communication for thousands of years. [1] In other words, bridges are remarkable structures that combine art, science, and practical purpose. From simple wooden footbridges to massive suspension bridges, each design solves specific challenges of span, load, and environment.

The key challenge is spanning distance safely. [2] A bridge must support its own weight plus the weight of traffic—cars, trains, or pedestrians—without bending or collapsing. Engineers choose different bridge types based on the distance to cross. Beam bridges work well for short spans; they rest on supports at each end. Arch bridges use a curved shape to distribute weight along the arch to the supports below—a design used since ancient Roman times. For very long spans, suspension bridges use cables hung between tall towers to hold the deck. The choice depends on the site, the materials available, and the budget.

Materials have evolved over time. [3] Early bridges were built from stone, wood, or bamboo. Today, steel and concrete are common—they are strong, durable, and can be shaped into complex forms. Reinforced concrete combines concrete's ability to resist crushing with steel's ability to resist stretching, making it ideal for many modern bridges. Engineers also consider corrosion（腐蚀）, earthquakes, and extreme weather. A bridge built to last decades must withstand forces that nature throws at it.

Building a bridge is a long and complex process. [4] Teams survey the site, build models, run computer simulations, and test materials before construction begins. Building over water often requires temporary structures—such as cofferdams（围堰）—to keep the work area dry. Large parts may be built elsewhere and transported to the site. Safety is paramount: every weld, bolt, and pour is checked. A single bridge can take years to complete and involve hundreds of workers, from engineers to welders to crane operators.

[5] Bridges are achievements of human ingenuity that serve us every day. The next time you cross a bridge—whether over a small stream or a wide river—take a moment to notice its design. Behind that structure are centuries of engineering knowledge and the hard work of countless people. Bridges remind us that with careful planning and cooperation, we can overcome the obstacles that divide us.

选项

A. In other words, bridges are impressive feats of engineering that serve practical needs. B. That is because different loads and spans require different structural designs. C. In fact, bridge materials and construction methods have advanced significantly over the centuries. D. These steps ensure that every bridge is built safely and to high standards. E. In short, bridges are symbols of human ability to connect and overcome obstacles. F. For example, the Golden Gate Bridge in San Francisco is one of the most photographed bridges in the world.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：引出桥梁作为工程杰作与实用设施。空前描述"Bridges connect places nature has separated...essential to human trade, travel, communication for thousands of years"，空后说明"bridges are remarkable structures that combine art, science, and practical purpose...From simple wooden footbridges to massive suspension bridges...each design solves specific challenges"。A选项"In other words, bridges are impressive feats of engineering that serve practical needs"（换言之，桥梁是服务实际需求的杰出工程成就）中"In other words"概括空前功能，"impressive feats of engineering...practical needs"与空后"remarkable structures...practical purpose"精准呼应，衔接顺畅。
- **[2] B**：
  - **线索**：不同荷载与跨度需要不同结构设计。空前提出"The key challenge is spanning distance safely"，空后详述"must support own weight plus traffic...without bending or collapsing...Engineers choose different bridge types based on distance...Beam bridges for short spans...Arch bridges...curved shape...distribute weight...suspension bridges for very long spans...cables...towers...choice depends on site, materials, budget"。B选项"That is because different loads and spans require different structural designs"（那是因为不同的荷载和跨度需要不同的结构设计）中"That is because"解释选型原因，"different loads and spans...different structural designs"与空后 beam、arch、suspension 等类型选择完美对应，逻辑紧密。
- **[3] C**：
  - **线索**：桥梁材料与施工方法的演进。空前提出"Materials have evolved over time"，空后列举"Early bridges: stone, wood, bamboo...Today: steel and concrete...strong, durable, complex forms...Reinforced concrete...resist crushing and stretching...consider corrosion, earthquakes, extreme

weather...withstand forces...last decades"。C选项"In fact, bridge materials and construction methods have advanced significantly over the centuries"（事实上，桥梁材料和施工方法在几个世纪以来取得了显著进步）中"In fact"强化演进，"materials and construction methods...advanced significantly"与空后stone/wood 到 steel/concrete、考虑腐蚀与抗震等精准对应，衔接自然。

- [4] D：
  - **线索**：施工流程确保安全与高质量。空前提出"Building a bridge is a long and complex process"，空后描述"survey site, build models, run simulations, test materials...over water: cofferdams...parts built elsewhere, transported...Safety paramount: every weld, bolt, pour checked...years to complete...hundreds of workers"。D选项"These steps ensure that every bridge is built safely and to high standards"（这些步骤确保每座桥梁都安全建造并符合高标准）中"These steps"指代空后调查、建模、模拟、检测、安全核查等，"safely and to high standards"与空后"Safety paramount...every...checked"完美对应，衔接得当。
- [5] E：
  - **线索**：全文收束与人类克服障碍的象征意义。空[5]位于结尾段段首，空后总结"achievements of human ingenuity...serve us every day...notice its design...centuries of engineering knowledge...hard work of countless people...with careful planning and cooperation, we can overcome obstacles that divide us"。E选项"In short, bridges are symbols of human ability to connect and overcome obstacles"（简言之，桥梁是人类连接与克服障碍能力的象征）用"In short"总结全文，"symbols...connect and overcome obstacles"与空后"overcome obstacles that divide us"形成呼应，收束精准。
- **干扰项分析**：
  - **F. For example, the Golden Gate Bridge in San Francisco is one of the most photographed bridges in the world.**：涉及金门大桥为最常被拍摄的桥梁，易被误填[1]或[5]。但F以"For example"引出金门大桥这一著名案例，而空[1]讨论的是桥梁的普遍功能与设计多样性，F的"most photographed"属于知名度与审美，与"combine art, science, practical purpose"中的工程论述焦点不同；空[5]总结桥梁作为人类智慧与克服障碍的象征，F的"photographed"侧重旅游与视觉，与"connect and overcome obstacles"的深层意义不符。F属"主题相关但论述层次错位"型干扰。